

INFECTION STRATEGIES OF *BOTRYTIS CINEREA* AND RELATED NECROTROPHIC PATHOGENS.

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1. Introduction

Botrytis cinerea Persoon: Fries (known as “grey mould fungus”) causes serious pre- and post-harvest diseases in at least 235 plant species (Jarvis, 1977), including a range of agronomically important crops, such as grapevine, tomato, strawberry, cucumber, bulb flowers and ornamental plants. Gramineous monocots are generally considered as poor hosts for grey mould. Disease control frequently relies on chemicals, although efforts to develop biological control strategies are increasingly successful (e.g. Köhl *et al.*, 1995; Elad, 1996) and biocontrol agents are marketed. The name of the asexual stage or anamorph, *Botrytis cinerea*, is preferred to the name of the teleomorph, *Botryotinia fuckeliana* (de Bary) Whetzel (XIth International Botrytis Symposium, 1996, Wageningen, The Netherlands). The teleomorph has rarely been detected in the field during the last century, but molecular population studies recently provided clear evidence that sexual reproduction occurs more frequently than previously anticipated (Giraud *et al.*, 1997). The pathogen is a typical necrotroph, inducing host cell death resulting in serious damage to plant tissues, culminating in rot of the plant or the harvested

product. There are extensive descriptions of microscopic and biochemical studies on infection mechanisms (reviewed by Staples and Mayer, 1995). Comprehensive insight in the infection process, however, is hampered by the fact that various groups used different fungal strains and different host species for their studies.

It is relatively recently that molecular-genetic tools such as transformation (Hamada *et al.*, 1994), differential gene expression analysis (Benito *et al.*, 1996) and gene cloning (van der Vlugt-Bergmans *et al.*, 1997a,b) have come available to unravel the factors involved in pathogenicity of *B. cinerea*. Application of these tools has provided substantial new data in a short time and it is expected to yield even more information in the next decade, which will be used to design novel strategies for grey mould control. Here we present an overview of the current insights and hypotheses on the mechanisms employed by *B. cinerea* to infect its host plants. The role of different fungal compounds in consecutive stages of the infection process is described. The balance between induced plant defense responses and the evasion of these responses by the fungus plays an important role in determining the outcome of the attempted invasion. Finally, we briefly discuss the analogy between infection mechanisms utilized by *B. cinerea* and related necrotrophs, which are as yet less accessible for molecular-genetic research.

2. Disease cycle

For the purpose of this review, different stages are distinguished in the disease cycle of *B. cinerea* (Figure 1). It should be emphasized that, given the wide host range of the fungus, not all processes occur in every infection.

B. cinerea can produce conidia on every host plant. They are ubiquitous in the air and can be transported by wind over long distances before infecting the next host (Jarvis, 1977). Following attachment, the conidium germinates under favorable conditions and produces a germ tube that penetrates the host surface. Whether true infection structures are produced during this process is a matter of debate. After surface penetration the underlying cells are killed and the fungus establishes a primary lesion, in which necrosis and defense responses may occur. In some cases this is the onset of a period of quiescence of an undefined length, in which fungal outgrowth is negligible (reviewed by Prusky, 1996). At a certain stage the defense barriers are breached and the fungus starts a vigorous outgrowth, resulting in rapid maceration of plant tissue, on which the fungus finally sporulates to produce inoculum for the next infection. Under optimal conditions, one infection cycle may be completed in as little as 3-4 days, depending on the type of host tissue attacked. In the following paragraphs we will discuss the various stages in the disease cycle and the role that enzymes and metabolites (mainly of fungal origin) play in these stages.

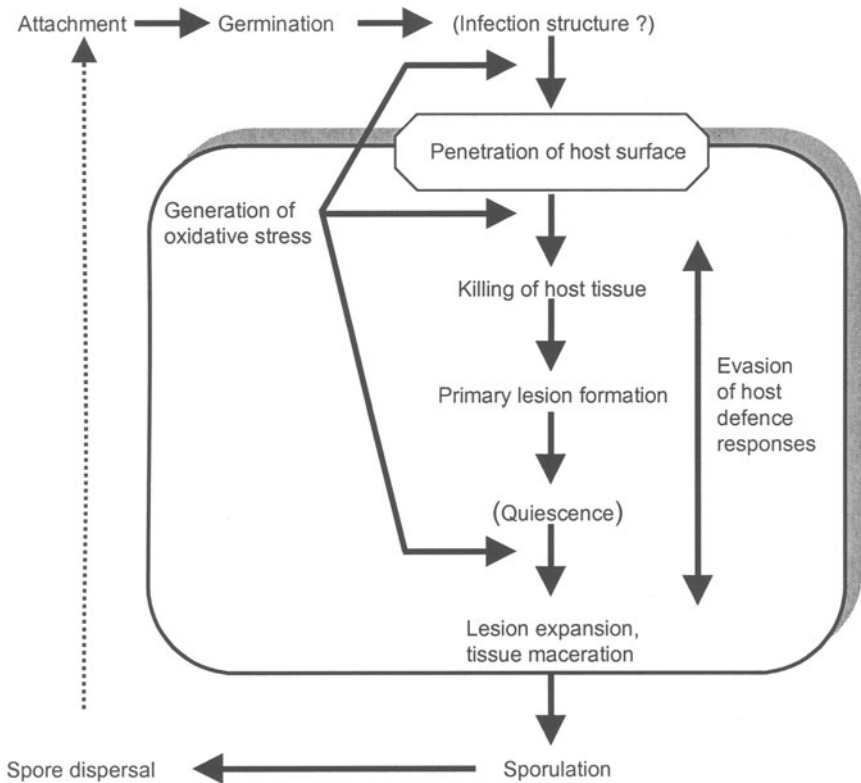


Figure 1: Different stages in the infection process of *B. cinerea*, to be discussed below. The shaded box represents the host tissue.

3. Attachment of conidia

3.1. DISPERSAL OF CONIDIA BY INSECTS

Before being able to penetrate host tissue the conidia of *B. cinerea* must land on and attach to the plant surface. The conidia are ubiquitous in the air and are capable of reaching any host by random chance. Specific trafficking by insects towards attachment sites of host tissue has also been described. The fruit fly *Drosophila melanogaster*, the New Zealand flower thrips *Thrips obscuratus*, as well as larvae of the grape berry moth *Lobesia botrana*, were shown to carry viable conidia of *B. cinerea* (Fermaud and Le Menn, 1992; Fermaud *et al.*, 1994; Fermaud and Gaunt, 1995; Louis *et al.*, 1996). Wounds inflicted on host tissue by the various insects increased disease development. The fruit fly not only carried conidia externally on its cuticle but also in the digestive tract, and microsclerotia were detected in the gut (Louis *et al.*, 1996).

3.2. SURFACE CHARACTERISTICS

A number of physical surface interactions is believed to play a role in the attachment of conidia to the plant epidermis. Conidia of *B. cinerea* have a hydrophobic surface which can usually only be wetted by solutions containing mild detergents. Williamson *et al.* (1995) described a granular structure on the surface of *B. cinerea* conidia, with a smooth area along the main axis. Doss *et al.* (1997) visualized numerous short (200-250 nm) protuberances on the rough surface of dry conidia of seven different *Botrytis* species. These protuberances disappeared upon hydration and redrying. The typical basket-weave pattern of hydrophobin rodlets present in *Magnaporthe grisea* (MPG1; Talbot *et al.*, 1996) was not observed on conidia of any of the *Botrytis* species tested (Doss *et al.*, 1997). Hydrophobins, which are involved in the interaction of *M. grisea* with the rice plant surface (Talbot *et al.*, 1993; Beckerman and Ebbole, 1996; Talbot *et al.*, 1996), thus seem to be absent in the *Botrytis* species studied, unless *Botrytis* hydrophobins aggregate in a structure different from *M. grisea* and other fungi (Kershaw and Talbot, 1998). Molecular research is needed to clarify whether or not *B. cinerea* possesses hydrophobin genes.

Doss *et al.* (1993, 1995) distinguished two steps in the attachment to host tissue. The first stage, preceding the hydration of conidia, is characterized by relatively weak adhesive forces, presumably resulting mainly from hydrophobic interactions between the host and conidial surfaces (Doss *et al.*, 1993). A stronger binding occurs in the second stage of adhesion, several hours after inoculation, when it becomes increasingly difficult to wash off the conidia. It should be noted, however, that at this stage conidia have developed into germlings. At this stage a fibrillar-like matrix material is present on the tip of germ tubes (Cole *et al.*, 1996; see Figure 2). The attachment of germlings to either hydrophobic or hydrophilic substrata coincides with the production of a base-soluble compound, forming a fibrillate sheath around the germlings (Doss *et al.*, 1995), supposedly consisting of an extracellular β -(1,3)(1,6)-*n*-glucan (cinerean). Cinerean forms a slimy adhering capsule in liquid culture. Once glucose is exhausted from the medium, *B. cinerea* degrades the polymer by cinereanase into glucose and gentobiose, which is subsequently hydrolyzed to glucose (Stahmann *et al.*, 1992; Stahmann *et al.*, 1993; Monschau *et al.*, 1997). Besides serving as glue and nutrient storage, cinerean may have a role in protecting hyphae against dehydration and defense mechanisms of the host.

When inoculation was carried out using conidia in aqueous glucose, a sheath of fibrillar-like material (presumably cinerean) was detected on the germ tube surface, whereas in inoculations with dry conidia this sheath is absent (Cole *et al.*, 1996). Since dry-inoculated conidia were also firmly attached to the plant surface, it was concluded that the sheath is not essential for attachment of conidia to the host.

4. Germination

After landing on the host tissue, a number of factors influence the germination of a conidium. Free surface water or high relative humidity (>93% RH) is required to germinate and penetrate the host epidermis (Williamson *et al.*, 1995). In addition, moisture assists the pathogen in the uptake of nutrients residing on the host epidermis or pollen grains (Blakeman, 1980). When dry conidia are inoculated on

plant surfaces and subsequently incubated in the absence of free surface water, the emerging germ tube usually remains shorter than the length of a conidium before it penetrates the surface (Salinas and Verhoeff, 1995; Williamson *et al.*, 1995; Cole *et al.*, 1996). Inoculation with conidia in an aqueous suspension, however, usually requires the addition of nutrients, which might mimic the situation in a wound on the plant epidermis, from which nutrients leach (Harper *et al.*, 1981; van den Heuvel, 1981). A highly efficient germination and synchronous infection of tomato leaves is obtained, when conidia are preincubated for 2-4 hours in liquid medium supplemented with phosphate and sugar (Benito *et al.*, 1998). The sugar is probably not only involved in stimulating the germination, but also in oxidative processes leading to host cell death (Edlich *et al.*, 1989; see paragraph 7).

Gaseous compounds might also be involved in the stimulation of germination. Elad and Volpin (1988) found a correlation between the level of ethylene production by flowers, petals and leaves of different rose cultivars, and the severity of grey mould symptoms. A stimulation of grey mould development by ethylene was also demonstrated in strawberry, tomato, cucumber and pepper (e.g. Elad and Volpin, 1988; McNicol *et al.*, 1989). This observation is usually ascribed to the weakening of the host tissue and senescence that coincides with ethylene production. The influence of ethylene on *B. cinerea* itself is only rarely investigated. Analogous to *Colletotrichum gloeosporioides* (Flaishman and Kolattukudy, 1994), germination of *B. cinerea* might be influenced by ethylene. Kepczynski and Kepczynska (1977) found that germination of *B. cinerea* conidia on a hydrophobic surface was stimulated by exogenous ethylene, but the germ tube length was unaffected. Application of 2,5-norbornadiene, a competitive inhibitor of ethylene perception in plants, inhibited germination in a reversible manner (Kepczynska, 1993). On the other hand, it was reported that in a hydrophilic environment, ethylene stimulated germ tube elongation without affecting the percentage of germination (Barkai-Golan *et al.*, 1989). It is tempting to speculate that ethylene produced by the plant during leaf senescence or fruit ripening might function as a signal for the conidia on the (hydrophobic) plant surface to germinate and initiate the infection. Subsequently the germ tube elongation might be stimulated by ethylene in the more hydrophilic environment of the invaded plant tissue. Thus, ethylene might favor grey mould development by weakening the host, as well as by stimulating germination of *B. cinerea* conidia and outgrowth of hyphae. Molecular and biochemical approaches are required to elucidate whether *B. cinerea* possesses ethylene receptors.

5. Differentiation of infection structures on the host surface

There is some debate whether *B. cinerea* forms appressoria during penetration. There is consensus that germ tubes do not differentiate into the highly organized appressoria that are typical for many plant pathogenic fungi (reviewed by Mendgen *et al.*, 1996). Several authors, however, reported an appressorium-like structure in *B. cinerea* but their nomenclature diverges. Van den Heuvel and Waterreus (1983) distinguished germ tube apices, *i.e.* tips of germ tubes appearing as an appressorium-like swelling, appressoria of different forms and infection cushions. Akutsu *et al.* (1981) grouped pre-penetration structures into primary, secondary or hyphal

elongations. It is probable that these structures, which were all observed upon inoculation with conidia in an aqueous glucose suspension, correspond to the structure shown in Figure 2, a swollen hyphal tip with an "adhesive pad" (Cole *et al.*, 1996). Secretion of a phosphatase by *B. cinerea* grown *in vitro* occurs specifically from vesicles in the hyphal tip (Weber and Pitt, 1997), which is typical for protein secretion in filamentous fungi (Sietsma *et al.*, 1995). It remains to be determined whether fungal hydrolytic enzymes, involved in host surface penetration, are also secreted at the hyphal tip and whether these enzymes accumulate prior to the onset of penetration, in the indented area delimited by arrows in Figure 2.

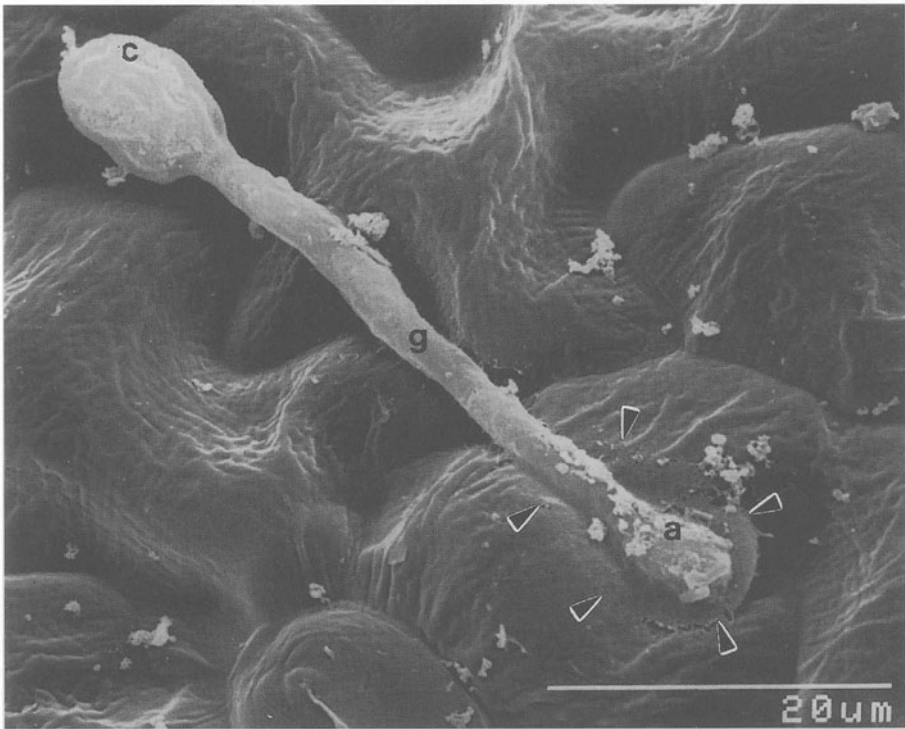


Figure 2: Scanning Electron Micrograph of an early stage of infection (5 h.p.i.) of tomato leaves. A conidium (c) has germinated and formed a germ tube (g), which terminates in a swollen tip (a) surrounded by matrix material, the adhesive pad. The host epidermal surface is partly indented around the site of contact with the adhesive pad (indicated by arrows). A stomatal cell is indicated (s).

The swelling of the hyphal tip may be the consequence of a rise in the osmotic value in the hyphal tip, resulting in water absorption. In the absence of a rigid layer in the outer wall, swelling can not result in an equally high turgor as is generated in appressoria of *M. grisea* (Howard *et al.*, 1991; de Jong *et al.*, 1997). The external sheath, presumably consisting of cinerean (see paragraph 3), may contribute to the swelling by retaining water, as cinerean is extremely hygroscopic. Whether the hydrophobic host surface or the release of volatile plant hormones provide signals to trigger the formation of hyphal swellings is purely speculative.

Swelling of hyphal tips is usually not observed when hosts are inoculated with dry conidia (Salinas and Verhoeff, 1995; Williamson *et al.* 1995; Cole *et al.*, 1996). In the latter cases, the conidia produced short germ tubes which directly penetrated the epidermal cells, subsequently leading to cell death.

6. Penetration of the host surface

Invasion of host tissue can be achieved by active penetration or passive ingress. *B. cinerea* is a renowned opportunist that can initiate infection at wound sites, or at sites that have previously been infected by other pathogens. *B. cinerea* can also enter the substomatal cavity via an open stoma. Only direct penetration of the epidermal surface is discussed in this paragraph. For reasons of simplicity the penetration of dead or wounded tissue, as well as via stomata, is regarded as an expansion rather than a penetration process, and is dealt with in paragraph 10.

6.1. CUTINASES

When conidia land on aerial parts of a plant, the first barrier to overcome is the cuticle, which covers the epidermal cells. Its major structural component, cutin, is a polyester composed of hydroxylated and epoxidized C₁₆- and C₁₈-fatty acids (Martin and Juniper, 1970). Physical damage or mechanical penetration of the cuticle by *B. cinerea* is not usually observed (Williamson *et al.*, 1995; Cole *et al.*, 1996). Hence, cutinolytic activity is presumably required to penetrate this layer (Salinas and Verhoeff, 1995; van der Vlugt-Bergmans, 1997a). Salinas (1992) investigated whether a particular 18 kDa cutinase is important in this process and raised monoclonal antibodies against the enzyme. Application of the antibody to gerbera flowers prior to inoculation reduced lesion formation by 80%. A cutinase-deficient gene replacement mutant, however, did not have any discernible reduction in virulence on gerbera flowers nor on tomato fruits, as compared to the wild type (van Kan *et al.*, 1997). Although the observations of Salinas (1992) remain to be explained, it can be ruled out that this particular 18 kDa cutinase is essential in penetration.

It should be taken into account that the 18 kDa cutinase, like all other cutinases studied thus far in plant pathogenic fungi, is most likely an exo-hydrolase. Enzyme activity of cutinases is usually defined by their ability to release soluble fatty acid monomers from the water-insoluble substrate (Purdy and Kolattukudy, 1973). It would be much more efficient for a pathogen to produce an endo-hydrolase, in order to create openings for penetrating a polymer by fungal hyphae. Such endo-cutinase, however, will not release water-soluble products from the insoluble cutin, since cleavage products most likely remain attached in the network. Hence, endo-cutinase activity is difficult to detect in biochemical assays.

One candidate for an enzyme with such activity is a 60 kDa lipase that is induced upon growth in liquid medium with apple cutin as the sole carbon source (Comménil *et al.*, 1998). This lipase possesses low but significant cutinolytic activity and it has clearly distinct kinetic properties than the 'typical' cutinases mentioned above. When polyclonal antibodies raised against this lipase were applied prior to inoculation with *B. cinerea* conidia, germ tubes were no longer able to penetrate the cuticle. The

antibodies did not affect germination (Comménil *et al.*, 1998). Whether the lipase plays an essential role in host tissue penetration should be assessed by cloning the corresponding gene, constructing a targeted lipase-deficient mutant and determining its virulence.

6.2. PECTINASES

Microscopic studies have shown that after penetration of the cuticle, hyphae of *B. cinerea* frequently invade the anticlinal wall between two epidermal cells. The concomitant swelling of the epidermal cell wall (Mansfield and Richardson, 1981) is indicative for the degradation of the pectin in the matrix of the epidermal wall, presumably as a result of water absorption, as will be discussed below in paragraph 7. Biochemical evidence suggested that pectinases might be involved in primary infection. At least one (basic) endopolygalacturonase (endoPG) is expressed constitutively and it was therefore proposed to be involved in early stages of the infection process (Van der Cruyssen *et al.*, 1994). Gene cloning revealed that *B. cinerea* contains an endoPG gene family, consisting of six members encoding basic as well as acidic isozymes (ten Have *et al.*, 1998; Wubben *et al.*, 1999a). Targeted deletion mutants were made in both genes encoding the basic endoPGs (*Bcpg1* and *Bcpg2*) by gene replacement. Both types of mutants were still able to cause primary necrotic lesions on non-wounded tomato and bean leaves (ten Have *et al.*, 1999), excluding an essential role for BcPG1 and BcPG2 in host surface penetration.

6.3. PROTEASES

Movahedi and Heale (1990a) detected extracellular aspartic protease (AP) activity in ungerminated conidia as well as during germination, prior to the appearance of pectinase activity. Application of the specific AP inhibitor pepstatin during inoculation markedly reduced infection of carrot slices, suggesting an important role for AP during primary infection (Movahedi and Heale, 1990b). Recently, a gene was cloned encoding an aspartic protease, *BcAP1*, and targeted mutants were made to study its involvement in the infection of detached tomato leaf tissue (Prins *et al.*, 1999a). No discernable loss of virulence was observed for the BcAP1-deficient mutant, indicating that this protease is not essential for virulence. Since *B. cinerea* probably contains at least one additional AP gene, the importance of aspartic proteases in pathogenesis can not yet be excluded.

7. Killing the host

B. cinerea kills host cells before they are invaded by hyphae (Clark and Lorbeer, 1976). Recent studies have demonstrated that invasion of plant tissue by *B. cinerea* triggers nuclear condensation and plant membrane damage, two indicators for programmed cell death, in a ring of cells around the hyphae (Govrin and Levine, unpublished results). These results imply that diffusible factors have a direct or indirect phytotoxic activity. Several phytotoxic compounds that have been proposed to play a role in killing host cells are discussed below.

7.1. TOXINS

Culture filtrates of *B. cinerea* may induce toxic effects when applied to plant tissue (Rebordinos *et al.*, 1996), but there is still little evidence for the existence of a causal fungal toxin *sensu strictu*. A *B. cinerea* isolate excreted in liquid culture a highly substituted lactone, botcinolide, which inhibited wheat coleoptile elongation and induced necrosis or chlorosis on bean, corn and tobacco plants (Cutler *et al.*, 1993). Homobotcinolide, a natural botcinolide homologue, did not exhibit phytotoxic effects on bean but it was more potent in inhibiting wheat coleoptile elongation (Cutler *et al.*, 1996). Four additional derivatives of botcinolides have been isolated and structurally characterized without elucidation of their biological activity (Collado *et al.*, 1996). The tricyclic sesquiterpenes, botrydial and dehydrobotrydial, were purified from liquid culture and were phytotoxic on tobacco leaf discs. However, both metabolites were secreted by *B. cinerea* only when the medium contained a high glucose level (>3%). Moreover the damage inflicted by the purified metabolites only occurred at concentrations significantly above the natural concentrations found in liquid media (Rebordinos *et al.*, 1996). Before one can envisage a significant role of phytotoxins of *B. cinerea* in pathogenesis, more information is needed on the timing of secretion, the occurrence of toxic metabolites *in situ* and their correlation with the aggressiveness of individual fungal strains.

7.2. OXALIC ACID

Secretion of oxalic acid (OA) is a widespread property of fungi from various taxonomic classes. Its occurrence and ecological function was recently reviewed by Dutton and Evans (1996). A key role for OA in pathogenesis has been postulated for several plant pathogens like *Sclerotinia sclerotiorum* (Godoy *et al.*, 1990), *Sclerotium rolsii* (Kritzman *et al.*, 1977), *Mycena citricolor* (Rao and Tewari, 1987), and *Sclerotium cepivorum* (Stone and Armentrout, 1985). The wide range of host species, tissue types and plant growth stages that are parasitized by some of these pathogens, has been ascribed to the phytotoxicity of OA. The sensitivity of bean cultivars to OA was closely correlated with the severity of infection by *Sclerotinia* white mould (Tu, 1985; Tu, 1989). Mutants of *S. sclerotiorum*, which are deficient in OA production, were unable to infect *Arabidopsis* plants (Dickman and Mitra, 1992) and the deficiency could be restored by supplementing the inoculum with OA (Godoy *et al.*, 1993). These studies do, however, not prove that OA acts solely and directly as a phytotoxin. It may simply be a component in a cascade of events leading to host cell death. This view is supported by other studies in which the damage induced by *Sclerotinia* could not be explained solely by the toxicity of OA (Callahan and Rowe, 1991; Rowe, 1993).

B. cinerea produces OA both *in vitro* (Gentile, 1954; Schroeder, 1972; Donèche *et al.*, 1985) and *in planta* (Verhoeff *et al.*, 1988). The sizes of lesions induced by several strains of *B. cinerea* on grapevine and bean leaves strongly correlated with the amount of OA secreted *in vitro* by these strains (Germeier *et al.*, 1994). However, in a time course study with inoculated bean leaf discs the amounts of OA, secreted at 14 hours post inoculation, when the first host cells collapsed, were insufficient to justify the tissue damage observed (Tiedemann *et al.*, unpublished). It is thus likely that OA is an important co-determinant of pathogenicity but not the

primary phytotoxic agent. In fact, the role of OA might be in a synergistic action with endoPGs during macerating plant tissue, as has been described in various plant pathogens (Bateman and Beer, 1965; Amadioha, 1993; Punja *et al.*, 1985; Stone and Armentrout, 1985). Many fungal endoPGs have an activity optimum at low pH (Rombouts and Pilnik, 1980) and are therefore stimulated by the simultaneous secretion of OA. In addition, OA is believed to enhance the pectin degradation resulting from endoPG activity, by sequestering the Ca^{2+} ions from (intact or partially hydrolyzed) Ca-pectates in the cell walls. The resulting calcium oxalate complex is insoluble and crystallizes in the infected leaf tissue (Figure 3). The removal of Ca^{2+} ions disturbs intermolecular interactions between pectic polymers and disrupts the integrity of the pectic backbone structure. Consequently, the pectic structure absorbs water and swells, as was described by Mansfield and Richardson (1981).

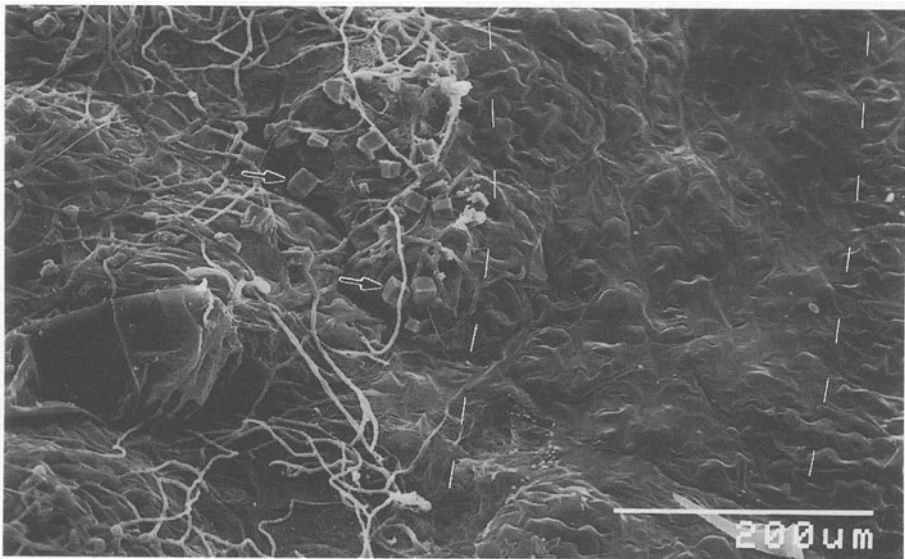


Figure 3: Formation of crystals (indicated by arrows) in tomato leaves infected by *B. cinerea*, visualized by Scanning Electron Microscopy. The white dashed line in the center of the picture indicates the border of the lesion between the external (ectotrophic) mycelium surrounded by crystals (on the left-hand side), and a concentric zone of collapsed epidermal cells (on the right-hand side). Both leaf zones are colonized by mycelium, growing below the epidermis. The white dashed line at the right-hand side of the picture represents the border between this endotrophically colonized area and the non-invaded leaf tissue.

7.3. INDUCTION OF ACTIVE OXYGEN SPECIES

Evidence that Active Oxygen Species (AOS) are involved in the attack by a pathogen, rather than in triggering plant defense, has been presented for several fungi, such as *Drechslera siccans*, *D. avenae* (Gönner and Schlösser, 1993), *Phytophthora infestans* (Jordan and DeVay, 1990) and *Cercospora beticola* (Daub and Hangarter, 1983). Studies on AOS production in relation to *B. cinerea* pathogenicity are more recent. Various antioxidants were found to reduce grey

mould disease development (Elad, 1992; Tiedemann, 1997). In a screening of a number of bean (*Phaseolus vulgaris*) genotypes, a correlation was found between the sensitivity of a particular genotype to oxidative stress and its susceptibility to *B. cinerea* (Tiedemann, 1997). Toxic AOS were detected *in situ* in bean leaf tissue infected with an aggressive strain of *B. cinerea*, and their appearance coincided with the occurrence of host tissue damage (Tiedemann, 1997). Transgenic tobacco plants, expressing an alfalfa ferritin, were more tolerant to oxidative reagents such as paraquat, and they were more resistant to infection by *B. cinerea* than the non-transformed control (Deák *et al.*, 1999).

One of the AOS produced at the host-fungal interface is hydrogen peroxide (H_2O_2 , see Figure 4). The level of H_2O_2 , released from bean leaf discs inoculated with different *B. cinerea* isolates, correlated with the aggressiveness of the isolate on such leaf tissue (Tiedemann, 1997). In *B. cinerea*-infected *Arabidopsis thaliana* leaves, H_2O_2 was detected in the apoplastic space as much as 5-10 cell layers away from the fungal hyphae (Govrin and Levine, unpublished results).

The mechanism by which *B. cinerea* induces (presumably toxic levels of) AOS in its host is still unclear. Fungal sugar oxidases might provide the source (Edlich *et al.*, 1989). It was proposed that the oxidative burst is a consequence of lipid oxidation by the invading fungus (Weigend and Lyr, 1996). Exogenous application of glucose oxidase or H_2O_2 , released during glucose oxidation, mimicked this phenomenon. The conclusion of Weigend and Lyr (1996) that a fungal glucose oxidase is responsible for the H_2O_2 accumulation at the host-fungal interface should be validated by molecular-genetic studies. Liu *et al.* (1998) purified and characterized a (presumably intracellular) glucose oxidase from mycelium of *B. cinerea*. More recently, a gene was cloned that encodes a different glucose oxidase, which contains a signal peptide and is therefore probably extracellular (Liu and Tudzynski, unpublished results). This gene will be used for gene replacement studies in order to study the role of extracellular glucose oxidase in pathogenicity.

Besides the production of AOS by fungal enzymes, it is probable that also the host contributes to a large extent to the generation of AOS, because of their extremely short half-life time. A prime source for generation of AOS is the plant 'oxidative burst' system, which consists of a plasma membrane-bound NADPH oxidase, inducible by fungal elicitors and requiring extracellular Ca^{2+} in the range of 1.5 mM (Schwacke and Hager, 1992). Infiltration of the specific NADPH oxidase inhibitor DPI into leaves of *Arabidopsis thaliana* prior to inoculation with *B. cinerea* resulted in a reduction of AOS production and a slower colonization of host tissue by the fungus (Govrin and Levine, unpublished results). Oligogalacturonides released from the plant cell wall by pectinases of *B. cinerea* are potential elicitors of an oxidative burst (Legendre *et al.*, 1993). This was demonstrated earlier in cotton leaf tissue, generating H_2O_2 in response to treatment with endoPG, in the presence of divalent cations (Mussell, 1973). Furthermore, OA may serve as a precursor of AOS in different ways, either through its enzymatic degradation by oxalate oxidase or by the chelation of ferric iron (Dutton and Evans, 1996). Although it is well established that oxidative damage is evoked by *B. cinerea* in the initial stages of attack, it remains to be clarified which factors determine the induction, generation and regulation of its phytotoxic components and how crucial this process is for penetration and colonization.

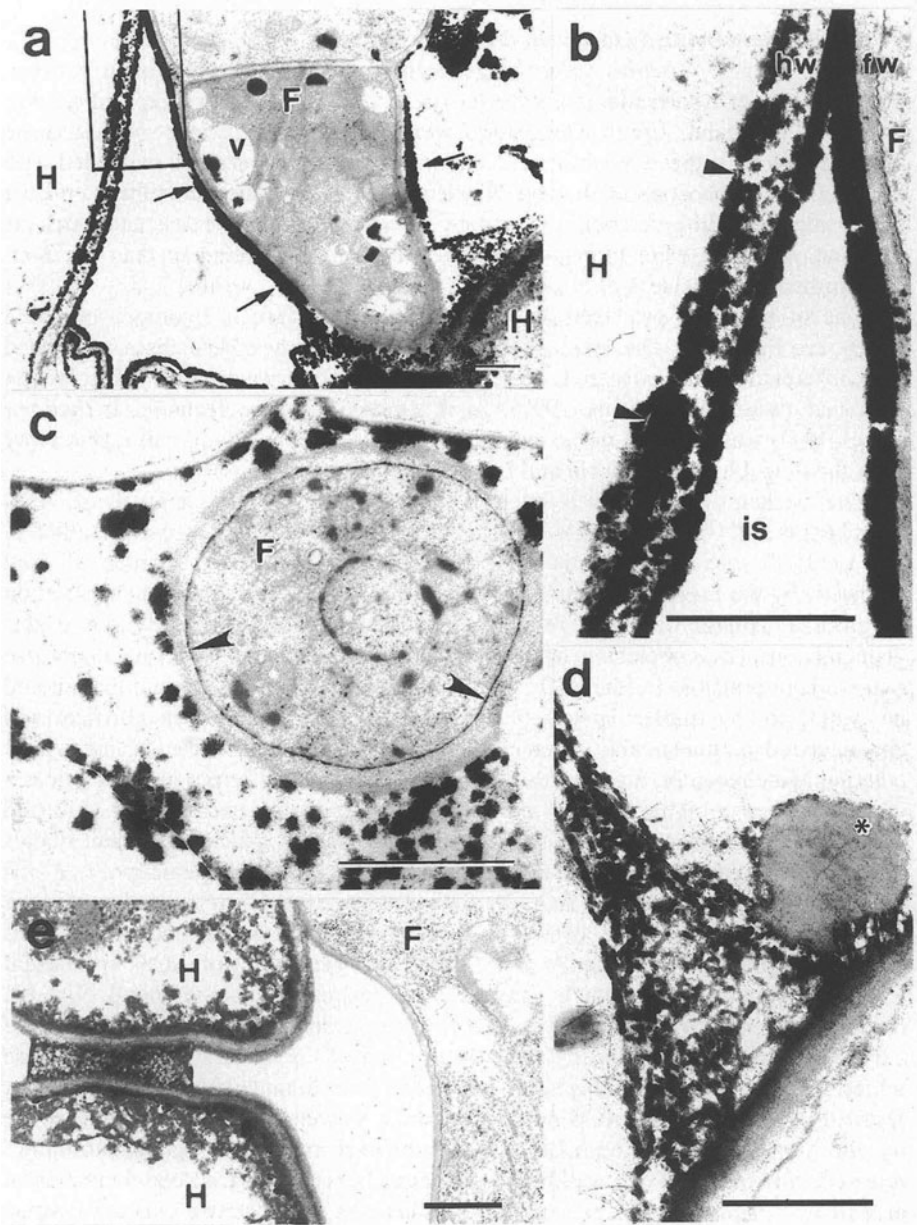


Figure 4: Localization of H_2O_2 in infected tomato leaf with cerium chloride (a-d) and visualization by Transmission EM. An electron-dense precipitate of cerium perhydroxide (indicated by arrows) is formed in the presence of H_2O_2 . (a), overview picture of the interface between host cells (H) and *B. cinerea* hyphae (F). A fungal vacuole is indicated by (v). (b), close-up of a section of (a), H_2O_2 is produced in the host cell (H) at the plasma membrane and diffuses through the host cell wall (hw) into the intercellular space (is). Electron-dense precipitate is observed in the periplasmic space (indicated by arrow-heads) and at the outside of the host cell wall (hw), as well as at the outside of the fungal wall (fw). The host cell is electron translucent and appears to be dead. (c), H_2O_2 production in hyphae. Electron-dense precipitate is observed inside a fungal cell that appears vital. (d), control, illustrating H_2O_2 production in lignifying xylem tissue. The asterisk indicates a cell wall thickening. (e), control of infected tissue that was not treated with cerium chloride. Panels a, b, c and e were stained 48 h.p.i, panel d was stained 24 h.p.i.

8. Formation of primary lesions, defense responses in the host

One potent defense mechanism of plants against invasion by pathogens is the deposition of lignin, occurring during hypersensitive response. A more limited response involves the deposition of callose at the site of penetration. Brett and Waldron (1996) proposed that disturbance of the plant plasma membrane, which maintains a calcium gradient across the membrane, results in a net calcium influx, which in turn triggers callose deposition by the calcium-dependent $\beta(1-3)$ glucan synthase complex. Both lignin and callose formation may put up a barricade that prevents further penetration of the pathogen into the protoplast.

The host surface penetration and the subsequent rupture of the plant cell walls by cell wall degrading enzymes (CWDEs) of *B. cinerea* triggers a cascade of processes in the fungus as well as the host. This paragraph deals with molecules, generated at the host-fungus interface, which trigger plant defense responses and with the impact of such responses on the progress of the infection.

8.1 ELICITORS

Among the plant cell wall breakdown products, cutin monomers may act as endogenous elicitors. Dihydroxypalmitic acid, the major component of potato cutin, is the active monomer that stimulates ethylene production, alkalinization, membrane depolarization, and transcriptional activation of defense-related genes (Schweizer *et al.*, 1996). Such elicitors may be released by the action of the 18 kDa cutinase discussed in paragraph 6. Fragments of host-derived pectic polysaccharides released by fungal pectinases may act as elicitors of different defense responses (Legendre *et al.*, 1993; Brett and Waldron, 1996).

In addition it was shown that several components released from fungal hyphae may serve as exogenous elicitors of defense responses in *in vitro* assays with cell suspensions like ergosterol (Granado *et al.*, 1995), chitin fragments (Felix *et al.*, 1993) and glycopeptides (Basse and Boller, 1992; Fath and Boller, 1996). The host may respond by transient alkalinization and membrane depolarization, coinciding with an oxidative burst.

8.2. INDUCTION OF NECROSIS

The initial establishment of primary necrotic lesions coincides with (and is in fact the result of) host defense activation in the neighboring tissue in response to the death of an invaded cell. It is as yet unclear whether cell death caused by a necrotroph, such as *B. cinerea*, is equivalent to cell death during a hypersensitive response (HR) to a biotrophic pathogen (reviewed by Lamb and Dixon, 1997). Recent studies by Govrin and Levine (unpublished results) have shown that an oxidative burst occurs in plant tissue several cell layers away from the fungal hyphae. Cytological staining provided evidence for rapid nuclear condensation and irreversible membrane damage, indicative of a programmed cell death process (Govrin and Levine, unpublished results).

Largely the same defense responses are activated during an infection by *B. cinerea* as during HR to avirulent races of a biotrophic pathogen: lignification (Maule and Ride, 1976; Heale and Sharman, 1977), biosynthesis of phytoalexins

(e.g. Bennett *et al.*, 1994), and PR proteins (e.g. Benito *et al.*, 1998). The total spectrum of defense responses results in a primary necrotic lesion in which the fungus is effectively restricted. Depending on the type of host tissue and yet unidentified physiological aspects of the host, the lesions enter a lag phase in which they become darker but do not expand. Only a proportion of the primary lesions is eventually able to develop into aggressive, expanding lesions that rapidly colonize the entire tissue (van den Heuvel, 1981; De Meyer and Höfte, 1997; Benito *et al.*, 1998). Even when only a small proportion of the primary lesions expands, the entire tissue can be destroyed within 24–48 hours. In the non-expanding lesions the fungus is not killed, since viable fungal mycelium could be recovered from all lesions (Benito, unpublished results).

Thus, an active defense contributes to restricting the fungus within the primary lesions. This was further emphasized by the work of Benito *et al.* (1998) who incubated inoculated tomato leaves at low temperature (4°C) and in darkness. Tomato is a rather cold-sensitive plant and its induced defense responses are limited and ineffective at 4°C. *B. cinerea*, however, is renowned for its ability to infect plant tissues, even when they are stored cold. Under these conditions the plant serves as a living albeit rather inert substrate, whereas *B. cinerea* can readily grow and has an advantage over the host. The primary penetration sites become water-soaked instead of necrotic, and all primary lesions develop into expanding lesions, often at fairly even growth rate (Benito *et al.*, 1998; ten Have *et al.*, 1998).

8.3. SYSTEMIC ACQUIRED RESISTANCE

It has recently been reported that an effective defense to *B. cinerea* may be pre-activated by a mechanism resembling Systemic Acquired Resistance (SAR, reviewed by Ryals *et al.*, 1996). De Meyer and Höfte (1997) demonstrated that bacterization of bean roots with a *Pseudomonas aeruginosa* strain resulted in a reduced number of expanding *B. cinerea* lesions on the leaves. The production of salicylic acid by the bacterium was required for triggering the SAR to *B. cinerea* (De Meyer and Höfte, 1997). Transgenic plants expressing a bacterial salicylate hydroxylase gene (NahG, Gaffney *et al.*, 1993) may be used to validate whether lesion restriction is mediated by salicylate signaling, providing further evidence for the importance of SAR in restricting *B. cinerea* lesion expansion. The potent grey mould biocontrol agent *Trichoderma harzianum* T39 (Elad *et al.*, 1994; O'Neill *et al.*, 1996) appears to act in part by conferring SAR (De Meyer *et al.*, 1998).

More recent studies have indicated that effective lesion restriction occurs in non-induced wild type *Arabidopsis thaliana* ecotypes. The ability to restrict *B. cinerea* lesion outgrowth was abolished in mutant *A. thaliana* genotypes carrying the *coi1* mutation (deficient in the jasmonate signaling pathway; Thomma *et al.*, 1998) and the *lsd1* mutation (involved in HR in gene-for-gene interactions; Govrin and Levine, unpublished results).

Even when lesion restriction occurs, the defense responses are eventually unable to stop *B. cinerea* colonization completely. As a necrotroph, *B. cinerea* is well adapted to surviving in a hostile environment in which toxic plant metabolites and antifungal proteins accumulate. Various screenings were performed *in vitro* to identify plant proteins that are capable of reducing growth of *B. cinerea*. Many plant defense proteins appeared to be potent growth inhibitors of *B. cinerea* like chitinase,

β -1,3-glucanase, and a class of low molecular weight proteins known as defensins (e.g. Broekaert *et al.*, 1989; Terras *et al.*, 1993; Chung *et al.*, 1997; Tabei *et al.*, 1997; Salzmänn *et al.*, 1998). Several of the corresponding genes have been or will be used in molecular resistance breeding, but no major success has yet been reported.

8.4. QUIESCENCE

In non-green tissues, *B. cinerea* often causes quiescent infections (reviewed by Prusky, 1996), in which no symptoms are discernible at first. Most prominent examples are soft fruits such as strawberry, raspberry and grape. In these hosts, *B. cinerea* frequently infects during the flowering stage and resides in the developing fruit tissue. No fungal growth occurs and the mycelium degenerates into fragments of irregular shaped cells. Fungal growth only resumes at the onset of fruit ripening. It has often been considered that high levels of fungitoxic or fungistatic compounds in immature fruits contribute to grey mould quiescence, since the level of these compounds decreases during the ripening process. Therefore, attempts have been undertaken to increase the levels of compounds of interest, or to prevent their degradation during ripening. The level of the stilbene phytoalexin resveratrol in grapes was shown to be correlated with grey mould resistance (Langcake, 1981; Bavaresco *et al.*, 1997). The effect of over-expressing stilbene synthase genes from *Vitis* in transgenic plants on the level of resistance towards *B. cinerea* was evaluated. A significant, partial resistance was obtained in tobacco (Hain *et al.*, 1993) but not in tomato (Thomzik *et al.*, 1997).

Besides phytoalexins, fruits usually contain high levels of proteinaceous inhibitors of fungal cell wall degrading enzymes, the PolyGalacturonase Inhibiting Proteins (PGIPs) and their level decreases during ripening. In view of this correlation and the significant role that polygalacturonases play in the infection (see paragraph 10), efforts to produce transgenic plants overexpressing PGIPs have been undertaken to obtain resistance towards *B. cinerea* (Graham *et al.*, 1996). This strategy has thus far met with limited success if any. One of the problems in this strategy is that PGIPs have a differential activity towards individual fungal endoPGs (Desiderio *et al.*, 1997). This makes it relevant to utilize PGIPs that are most potent against the *B. cinerea* endoPG isozymes that are important in virulence (see paragraph 10).

9. Evasion of chemical defense

Plant pathogenic fungi have developed various mechanisms to overcome the deleterious effect of preformed (phytoanticipins) or induced (phytoalexins) chemical defense agents. The major strategy is an enzymatic detoxification of these compounds (reviewed by VanEtten *et al.*, 1995; Osbourn, 1996; Osbourn *et al.*, 1998). Although the ability to detoxify these chemicals is widespread among phytopathogens, an essential role for the successful colonization of a host has so far only been proven in a few cases (reviewed by Tudzynski and Tudzynski, 1998).

Because of its broad host range, *B. cinerea* encounters a wide spectrum of antimicrobial compounds synthesized by the various host plants. The ability of *B.*

cinerea to degrade or detoxify phytoalexins was intensively studied already over two decades ago (see review of the "older" literature by Mansfield, 1980). In some cases this ability obviously contributed to the virulence, *e.g.* in *Capsicum frutescens* (capsidiol, Stoessel *et al.*, 1972) and *Phaseolus vulgaris* (phaseollin, van den Heuvel, 1976). *B. cinerea* is also able to degrade the phytoalexins rishitin, medicarpin and maackiain, but this ability did not directly influence its ability to infect the respective host plants (Mansfield, 1980).

The best studied example for phytoalexin detoxification by *B. cinerea* that correlates with virulence, is the detoxification of the *Vitis* phytoalexins pterostilbene and resveratrol. The ability of fungal isolates to detoxify these phytoalexins was correlated to their virulence (Sbaghi *et al.*, 1996). Conversely, the resistance level of *Vitis* genotypes against grey mould is correlated to their phytoalexin content (Langcake, 1981; Jeandet *et al.*, 1992). *B. cinerea* produces a substrate-specific laccase (stilbene oxidase) that is able to oxidize both compounds to non-toxic derivatives (Pezet *et al.*, 1991). The enzyme has been purified and characterized (Pezet, 1998) and the mechanism of stilbene degradation has been elucidated in detail (Breuil *et al.*, 1998). The elicitation of stilbene biosynthesis in *Vitis* plants by a soil bacterium can be used as a biological control mechanism against *B. cinerea* infection of grape (Bernard *et al.*, 1998). Thus, in spite of the capability of *B. cinerea* isolates to degrade the *Vitis* phytoalexins, these compounds remain effective chemical defense compounds. The exact role of stilbene oxidase in pathogenicity of *B. cinerea* remains to be validated by a molecular-genetic approach, by cloning and deletion of the stilbene oxidase gene.

B. cinerea is also able to detoxify preformed antimicrobial compounds. Verhoeff and Liem (1975) showed that *B. cinerea* can degrade the tomato saponin α -tomatine, and suggested that this ability is correlated with the resistance against this phytoanticipin and with latency on tomato. Recently, Quidde *et al.* (1998) analyzed the degradation of α -tomatine by a field isolate of *B. cinerea* in detail. They showed that the saponin is not completely deglycosylated (as suggested by Verhoeff and Liem, 1975), but that only the terminal xylose is removed, yielding β_1 -tomatine. β_1 -tomatine appeared to be far less toxic than α -tomatine, confirming that this deglycosylation step represents a detoxification. The corresponding enzyme, a "tomatinase", was purified and characterized biochemically. A field survey showed that most *B. cinerea* isolates tested (from various host plants and geographic origin) possessed tomatinase activity, with only one exception (Quidde *et al.*, 1998). Interestingly, the strain lacking tomatinase activity was highly sensitive to α -tomatine, completely non-pathogenic on tomato, but highly aggressive on *Phaseolus*, strongly suggesting that the ability to detoxify α -tomatine is correlated with the virulence of *B. cinerea* on tomato. However, this strain might have further defects that cause the specific loss of virulence on tomato; a disruption of the tomatinase gene will be necessary for a final proof.

B. cinerea is also able to detoxify saponins other than α -tomatine. Quidde *et al.* (1999) demonstrated that the fungus can deglycosylate digitonin, avenacin, and avenacosides. They purified an "avenacinase" and showed that it is highly specific to avenacin and has no side activity, *e.g.* against avenacosides. Deletion of a putative "saponinase" gene (*sap1*) led to loss of avenacinase activity only. The role of an "avenacinase" for virulence of *B. cinerea* is open, since avenacin has so far only been detected in *Avena* roots, which are not infected by *B. cinerea*. Probably

comparable saponins occur in other plants that are potential hosts for the fungus. Taken together, the available data demonstrate that *B. cinerea* synthesizes at least three different saponinases: two glucosidases (avenacinase/avenacosidase) and a xylosidase (tomatinase, digitoninase). Since such analyses are limited by the availability of substrates, it is very likely that *B. cinerea* possesses an even larger set of enzymes for the effective detoxification of phytoanticipins.

In addition to substrate-specific detoxifying enzymes, a less specific enzyme might be involved in counteracting the effect of plant defense compounds that inhibit the growth of *B. cinerea*, namely glutathione S-transferase (GST). This enzyme has been studied especially in mammals and in plants in the past few years (Marrs, 1996). GST is able to conjugate glutathione to toxic compounds that accumulate in the cytoplasm. The resulting conjugate is subsequently transported to the vacuole or secreted. Many organisms contain a set of GST isozymes, which accept a variety of toxic substrates, including xenobiotics, phenolics, heavy metals and H₂O₂. A GST gene was cloned from *B. cinerea* (Prins *et al.*, 1999b), which is present in a single copy in the genome. A targeted gene disruption was performed and the resulting mutant was tested for a possible loss of virulence on tomato leaves. The mutant was still pathogenic, indicating that the GST gene is not essential for infecting this host tissue (Prins *et al.*, 1999b). In view of the wide substrate range of GSTs, it is difficult to predict against which toxic compounds the *B. cinerea* GST is supposedly conferring protection, and whether this has consequences for the infection on particular host plants.

Besides detoxifying antifungal chemicals that accumulate in plant tissue, a second mechanism has been proposed for fungi to overcome the growth inhibiting effects of these compounds. Energy-dependent secretion by ABC-transporters was postulated to provide plant pathogenic fungi with a tool to prevent the fungistatic or fungitoxic effects of such compounds (de Waard, 1997). ABC transporters have been cloned from *B. cinerea* and their role in resistance towards plant defense compounds and virulence on various hosts is being studied (Del Sorbo, Schoonbeek and de Waard, unpublished results). Some ABC-transporters are located in the vacuolar membrane and are involved in pumping complexes of glutathione and toxic compounds, generated by GST, into the vacuole (Ishikawa *et al.*, 1997).

The oxidative burst that occurs at the host-pathogen interface (see Figure 4 and paragraphs 7 and 8) imposes an oxidative stress on the host as well as the pathogen. During the infection on tomato leaves, the host catalase mRNA level increased from the moment of necrotic lesion appearance (van der Vlugt-Bergmans *et al.*, 1997b). The pathogen needs to be able to cope with external oxidative stress in order to survive in the necrotic tissue. Successful detoxification of extracellular AOS by *B. cinerea* was indicated by the observation that an intracellular catalase of *B. cinerea* is not expressed *in planta* in necrotic tissue, in spite of the fact that it can be induced *in vitro* in liquid culture by exogenously applied hydrogen peroxide (van der Vlugt-Bergmans *et al.*, 1997b). The removal of potentially toxic external AOS by *B. cinerea* is likely to be mediated by extracellular enzymes, such as Superoxide Dismutase (SOD) and catalase. Genes encoding SOD (Quidde, Weltring and Tudzynski, unpublished results) and extracellular catalase (Schouten and van Kan, unpublished results) have been cloned, and are currently characterized. If SOD and extracellular catalase are involved in self-protection of the pathogen against the oxidative stress that it imposes on the host, targeted replacement of the

corresponding genes in *B. cinerea* might yield a mutant that is killed by the oxidative burst occurring within the necrotic lesion.

Taken together, the biochemical and preliminary molecular-genetic data demonstrate the enormous metabolic versatility of *B. cinerea*, enabling the fungus to overcome a wide spectrum of antifungal plant defense mechanisms. Whether this metabolic capability contributes to the broad host-range of the fungus is under investigation in various laboratories.

10. Disease expansion and tissue maceration

Besides needing to cope with host defense responses, as described in paragraph 9, *B. cinerea* must be able to macerate plant tissue and utilize it for its own growth. The initial step in expansion of primary lesions is presumably the killing of neighboring cells by mechanisms similar to the ones described in paragraph 7.

In order to grow out of the primary lesion, the plant tissue must be actively degraded. First, the lignified barrier surrounding the primary lesion may need to be breached. It has not yet been studied how this process is achieved, but it is likely to involve peroxidases and other enzymes with lignin degrading capacity. It was proposed that laccase activity secreted by *B. cinerea* can interfere with plant defense responses by oxidizing phenolic compounds that serve as lignin precursors or as phytoalexins (Viterbo *et al.*, 1992). Moreover, some fungal laccases are capable of degrading lignin (Evans and Betts, 1991). Cucurbitacins, a group of compounds from Cucurbitaceae, inhibited the production of extracellular laccase activity by *B. cinerea* and conferred resistance to the pathogen (Bar Nun and Mayer, 1990).

Once through the lignin barrier, access to the surrounding tissue is obtained and this tissue is rapidly colonized. Cell wall degradation facilitates the entry of the pathogen and it provides nutrients for growth. For this purpose, *B. cinerea* possesses a set of cell wall degrading enzymes (CWDEs). The enzymes are produced during all stages of infection. Among the CWDEs of *B. cinerea* described by various groups are one or more pectin lyases (Movahedi and Heale, 1990b), pectin methylesterase (Reignault *et al.*, 1994), endopolygalacturonase (Johnston and Williamson, 1992), exopolygalacturonase (Johnston and Williamson, 1992) and cellulase (Barkai-Golan *et al.*, 1988).

It has been reported by several groups that treatment of plants with calcium fertilizer contributes to inhibition of grey mould development (Elad and Volpin, 1988; Wisniewski *et al.*, 1995; Chardonnet *et al.*, 1997; Klein *et al.*, 1997), although the effect may be tissue dependent. In cucumber, calcium treatment was able to reduce disease, by reduction of enzymatic pectin degradation on fruits but not on leaves (Chardonnet and Donèche, 1995). It is most likely that the protective effect of calcium is caused by the reinforcement of Ca-pectate complexes into egg-box structures, which are less accessible to CWDEs.

Recently, a number of *B. cinerea* genes have been cloned that encode CWDEs: pectin lyase (Mulder, unpublished), rhamnogalacturonan-hydrolase (Chen *et al.*, 1997) and six genes encoding endoPGs (ten Have *et al.*, 1998, Wubben *et al.*, 1999a). The endoPG genes, denoted *Bcpg1-6*, constitute a well studied and, most probably, complete gene family. The expression patterns of the individual endoPG genes *in planta* depend on the host tissue that is infected, on the progress of

infection, as well as on the external conditions at which infection occurs (ten Have *et al.*, 1999). Expression patterns *in planta* are in accordance with the expression data obtained in liquid cultures (Wubben *et al.*, 1999b) and they can largely be explained by combinations of inducing and repressing conditions in the host tissue. *Bcpg1* is expressed constitutively in all tissues and on all carbon sources tested. Other genes (*Bcpg4* and 6) are induced by the end product of PG activity, galacturonic acid, while the *Bcpg3* gene is induced at low pH and the *Bcpg5* gene is induced by growing on apple pectin. These studies suggest that *B. cinerea* can degrade pectate under different environmental conditions. It may be argued that there is a relation between the host range and the need to possess such a large gene family. The related plant pathogenic fungus *Sclerotinia sclerotiorum* also has a substantial endoPG gene family, as well as a broad host range (Fraissinet-Tachet *et al.*, 1995). However, Southern analysis showed that homologues of all members of the *B. cinerea* endoPG gene family are present in *Botrytis* species that can only infect a single host plant species (Wubben *et al.*, 1999a). This observation makes it unlikely that the mere presence of a family of endoPG genes is sufficient to explain the broad host range of *B. cinerea*, as compared to other *Botrytis* species.

Two genes encoding endoPGs (*Bcpg1* and *Bcpg2*), were eliminated by gene replacement and both mutants were reduced in virulence on tomato as well as on broad bean leaves (ten Have *et al.*, 1998; ten Have *et al.*, 1999). The effect of the deletion was most pronounced for the *Bcpg1* gene. This gene is expressed constitutively and could therefore be important in facilitating intercellular growth at the outer edge of the diseased tissue. Another explanation for the reduction in virulence may be purely nutritional. The absence of one enzyme will hamper the release of pectin degradation products that serve as nutrients, consequently resulting in slower growth of the fungus through the tissue. These studies clearly show that *B. cinerea* CWDEs can be involved in the invasion of host tissue surrounding the primary infected area. In various other plant pathogenic fungi, targeted deletion of CWDE genes resulted in strains with no discernable loss of virulence (Schaeffer *et al.*, 1994; Bowen *et al.*, 1995; Sposato *et al.*, 1995; Apel-Birkhold and Walton, 1996; Gao *et al.*, 1996; Görlach *et al.*, 1998; Scott-Craig *et al.*, 1998). The only other fungus in which a significant role of an endoPG in pathogenesis has been reported, is the saprophyte *Aspergillus flavus*, where endoPG deletion resulted in a reduced ability to invade cotton bolls (Shieh *et al.*, 1997).

We hypothesize that *B. cinerea* should be regarded as a "pectolytic" fungus. The host range of *B. cinerea* is confined to dicotyledons and non-graminaceous monocotyledons. All these hosts have a rather high content of pectin in the cell walls, when compared to the graminaceous non-host plants (Carpita and Gibeaut, 1993). In view of the role of endoPGs in virulence of *B. cinerea*, a suitable control strategy might be based on the use of PGIPs, which can inhibit endoPGs from *B. cinerea* (Johnston *et al.*, 1994). Therefore, PGIP genes may be used in molecular resistance breeding (Graham *et al.*, 1996), taking into account the differential activity of PGIPs towards individual fungal endoPGs (Desiderio *et al.*, 1997).

11. Phytohormone production by *B. cinerea*

Changes in the level of one individual plant hormone, or in the balance between different hormones, cause developmental changes in the plant which may result in symptoms such as stunting, overgrowth, leaf curls, and can even lead to plant death. Many of these symptoms are similar to those caused by plant pathogenic fungi. Interestingly, some plant pathogenic fungi are able to synthesize phytohormones in culture. Infected plant tissue often contains higher levels of certain phytohormones than uninfected plants (Isaac, 1992). However, it is unclear whether the hormone increase originates from the plant or from the fungus, whether pathogens influence the plant-directed hormone synthesis, or whether external application of hormones influences the disease. These knowledge gaps hamper a clear evaluation of the role of plant hormones in disease etiology.

B. cinerea is often regarded as a pathogen that mainly attacks weakened or senescing tissues. Hence, it was logical to determine the effects of applying exogenous hormones that stimulate plant senescence or ripening. Increased susceptibility to grey mould was shown to be associated with factors that enhance aging of the host tissues, such as abscisic acid (ABA) and ethylene (Elad, 1988; McNicol *et al.*, 1989; Elad and Evensen, 1995). ABA was also found to be a strong inhibitor of the accumulation of the phytoalexins rishitin and lubimin in potato tuber. The disease development on rose flowers was promoted by application of ABA, although germination and germ-tube elongation of *B. cinerea* were unaffected by ABA (Shaul *et al.*, 1996).

B. cinerea is able to produce several plant hormones (see Tudzynski, 1997). ABA is one of the naturally occurring phytohormones produced by *B. cinerea* (Marumo *et al.*, 1982; Kettner and Dörffling, 1995; Wu and Shi, 1998) and other phytopathogenic fungi such as *Cercospora rosicola* (Assante *et al.*, 1977), *Ceratocystis*, *Fusarium* (Dörffling and Petersen, 1984; Tuomi *et al.*, 1995) and *Alternaria brassicae* (Dahiya *et al.*, 1988).

Among 95 strains of *B. cinerea* tested, 39 produced ABA in axenic culture at a yield up to several mg/l (Hirai *et al.*, 1986). Kettner and Dörffling (1995) determined the levels of ABA in tomato leaves infected with ABA producing or non-producing *B. cinerea* strains and concluded that at least four processes control the level of ABA: 1. Release of ABA or its precursor by the pathogen; 2. Stimulation of ABA biosynthesis in the host plant; 3. Inhibition of plant ABA metabolism; 4. Promotion of biosynthesis of fungal ABA by the host tissue.

The ABA level in tomato leaves increased after infection, both with an ABA-producing and a non-producing *B. cinerea* isolate, albeit much more in response to the ABA-producing strain (Kettner and Dörffling, 1995). This indicates that ABA favors infection, but the ability of the fungus to produce ABA is not a prerequisite for infection of tomato leaves. In studies on the interaction of ABA-producing *B. cinerea* isolates with *Salix* leaves, a positive correlation was found between the frequency of infection and the amount of ABA produced in the infected tissue (Tuomi *et al.*, 1993). However, the ABA quantities detected in the tissue were too high to be secreted by the fungus, and presumably resulted mainly from increased ABA production by the plant.

Several *B. cinerea* strains isolated from infected fruits are able to produce ethylene in liquid culture. The ethylene levels are physiologically significant,

suggesting that ethylene produced by *B. cinerea* itself may play a role in the interaction between host and pathogen (Qadir *et al.*, 1997).

Cytokinin-like substances with cytokinin activity were detected in culture fluids of pathogenic *B. cinerea* isolates and other pathogens (Talieva and Filimonova, 1992). Since exogenous application of cytokinin causes abnormalities in plants such as tumor formation, overgrowth, galls and rusts, a possible involvement of fungal cytokinins in development of plant infection was suggested.

A systematic study of phytohormone production by *B. cinerea* and its role in pathogenicity is only in its early stages. It is difficult to determine whether the observed effects are due to growth regulators produced by the invading fungus itself, or by the plant in response to the stress evoked by the pathogen. Cloning of the biosynthetic genes of *B. cinerea* phytohormone pathways, followed by gene expression and gene replacement experiments are needed to resolve the role of hormone production by the fungus in pathogenesis.

12. Other *Botrytis* species and other necrotrophs

12.1. *BOTRYTIS* SPECIES WITH A NARROW HOST RANGE

In contrast to *Botrytis cinerea* that can infect at least 235 plant species, virtually all other (± 20) pathogenic *Botrytis/Botryotinia* species have a narrow host range, which is restricted to one or a few species of the corolliferous monocotyledons, the *Ranunculaceae* or the *Leguminosae* (Jarvis, 1977). The symptoms and development of diseases caused by the other *Botrytis* species on their host are often similar to those caused by *B. cinerea*, i.e. primary necrotic lesions of which a proportion subsequently expands into healthy tissue, thereby destroying large parts of the plant. Most of the infection mechanisms of narrow host range *Botrytis* species are probably similar to the mechanisms described above for *B. cinerea*.

Epidemics of *B. elliptica* in lilies and *B. tulipae* in tulips can occur simultaneously in regions in The Netherlands where tulips and lilies are co-cultivated. When *B. elliptica* is inoculated onto the non-host tulip, it is able to cause primary necrotic lesions that are barely distinguishable from those caused by *B. tulipae*. The lesions of *B. elliptica*, however, do not expand on tulip whereas those of *B. tulipae* do. The reciprocal is true for *B. tulipae* on the non-host lily. *B. cinerea* frequently occurs as a secondary opportunistic pathogen on tulip and lily leaves previously invaded by either of the other *Botrytis* species.

The determinants of the limited host range of *Botrytis* species, other than *B. cinerea*, are yet unclear. Specificity is apparently not determined by recognition of the host or signal exchange needed to establish an infection. In many cases, host-specific *Botrytis* species are able to cause primary lesions on a non-host, but these primary lesions fail to expand. The decisive step whether a plant is regarded as a host or a non-host thus lies in the lesion expansion. The effective restriction of the pathogen within the primary lesion needs to be breached, by neutralizing the fungitoxic and fungistatic (phenolic) metabolites that have accumulated within the lesion and by breaking through the fortified lignin barriers surrounding the lesion.

The broad host range of *B. cinerea* might thus be a consequence of its ability to neutralize defense mechanisms of a large number of host species (as discussed in

paragraph 9), whereas other *Botrytis* species are possibly only able to neutralize the defense mechanisms of one particular plant species, *i.e.* their specific host. Only very few studies have been reported on the ability of host-specific *Botrytis* species to neutralize fungitoxic compounds from their respective hosts.

As discussed in paragraph 9, Quidde *et al.* (1998) characterized a tomatinase-deficient *B. cinerea* field isolate that was non-pathogenic on tomato, yet retained the ability to infect bean as any other isolate. This suggests that host specialization or preference may occur even within the *B. cinerea* population. There is recent evidence that genetically isolated, sexually reproducing, sympatric populations (denominated *transposa* and *vacuma*) co-exist within the species *Botryotinia fuckeliana* in the Champagne region (Giraud *et al.*, 1997). One of the populations appeared to be more stable over time than the other and it was proposed that this represented a resident population, which was better adapted to the host plant(s) in the area (Giraud *et al.*, 1997). It will be highly interesting to investigate whether separate fungal populations have a preference for certain (single or groups of) host plant species. If such subgroups are indeed specifically adapted to subsets of host plants, the statement that *B. cinerea* is able to infect at least 235 host species (Jarvis, 1977) may represent an oversimplification.

12.2. *SCLEROTINIA SCLEROTIURUM*

Sclerotinia sclerotiorum is a wide host range pathogen with many similarities to *B. cinerea*. Both fungi belong to the family Sclerotiniaceae. The major difference is that *S. sclerotiorum* ("white mould") does not normally produce asexual spores. In the field it usually infects the base of the stem by ascospores released from apothecia that develop in the top layer of the soil in the spring. There are probably many things in common among the infection mechanism utilized by the two pathogens. Alike *B. cinerea* (see paragraph 10), *S. sclerotiorum* secretes a spectrum of CWDEs with differential regulation patterns (Fraissinet-Tachet *et al.*, 1995; Fraissinet-Tachet and Fèvre, 1996; Martel *et al.*, 1996). Whether these enzymes play a role in the infection process could thus far not be validated, since gene replacement has not yet been achieved in *S. sclerotiorum*. Mutants of *S. sclerotiorum* that are deficient in producing oxalic acid (OA) are significantly less virulent than the wild type and the loss of virulence can be restored by adding exogenous OA in the inoculum (Godoy *et al.*, 1993). The genes responsible for the OA deficiency in *S. sclerotiorum* have not been identified. It is likely that OA has an analogous role in pathogenesis of *B. cinerea* (see paragraph 7), but final validation of this hypothesis should come from the construction of an OA-deficient *B. cinerea* mutant.

13. Summary and concluding remarks

Tools have come available to start validating the numerous microscopic and biochemical observations that have been made on the infection strategies of *B. cinerea*. Figure 5 recapitulates our current view on the separate stages of the infection process and summarizes the involvement of various fungal enzymes and metabolites in these stages, as discussed in this chapter. Some of the corresponding genes have been analyzed in recent years, others will soon follow.

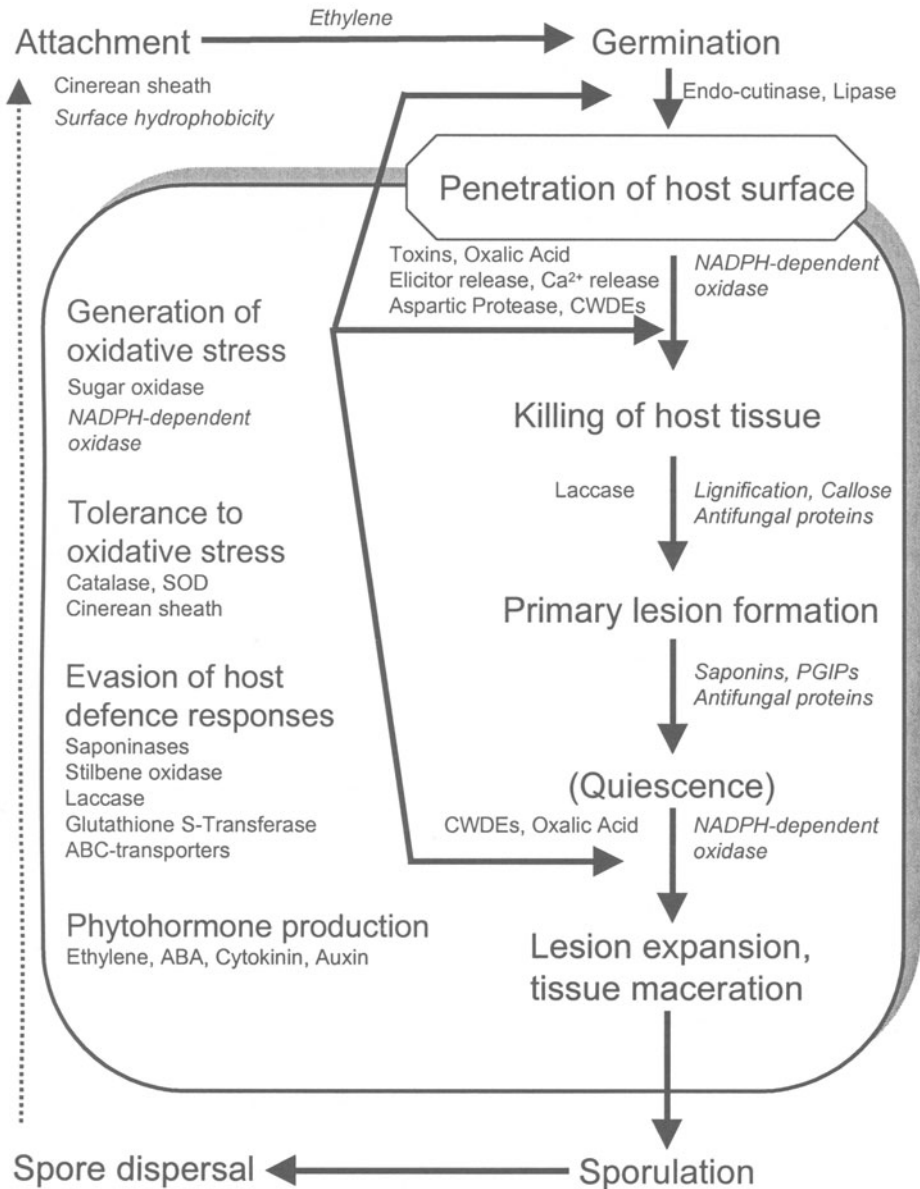


Figure 5: Schematic model of the involvement of various enzymes and metabolites in different stages of the infection process of *B. cinerea*. The shaded box represents the host tissue. Compounds produced by the pathogen are indicated in normal font, compounds originating from the host plant are indicated in italics.

The major challenge is in integrating the various factors in a comprehensive scheme of events. Such knowledge will be useful for developing new, rational disease control strategies.

Botrytis cinerea is a complex pathogen that can probably not be controlled by simple measures. This review demonstrates that the pathogen is versatile and uses a combination of factors during pathogenesis. In spite of the fact that necrotrophs appear to avoid the (usually subtle) interactions with their host that are typical for biotrophs, there seems to be a delicate balance between the attack mechanisms of the fungus and the defense of the host. Rather than combating the fungus vigorously by one single approach, it should be attempted to alter this balance in favor of the plant. Such a strategy will likely consist of a sophisticated combination of biological control agents, appropriate (partially) resistant plant genotypes and chemicals that either enhance the plant defense response or interfere with crucial steps in the infection process.

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